4.CCNP Enterprise作业四——STP 混合实验

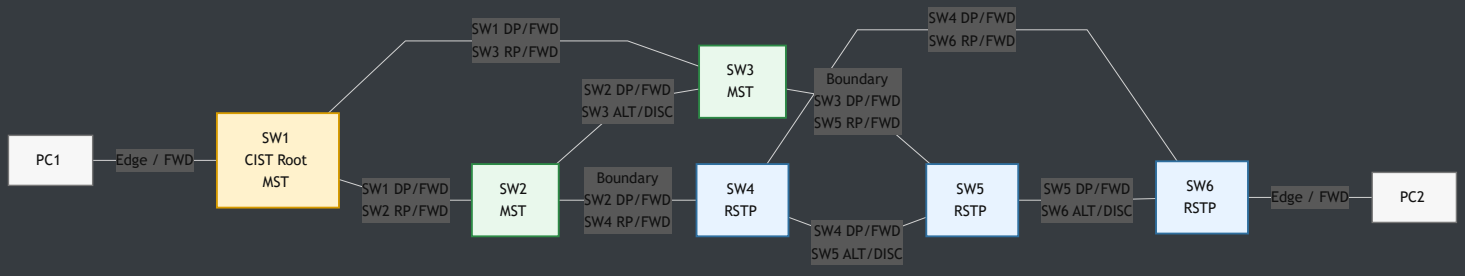
作业主题

MST 与 RSTP 混合环境

作业目标

1. 理解 MST、RSTP 与 CIST 的关系
2. 掌握 MST Region 的基本规划与配置
3. 掌握 MST 与 RSTP 混合环境中的边界端口识别
4. 理解 Root Port、Designated Port、Alternate Port 的选举结果
5. 验证 STP/RSTP/MST 的防环与快速收敛效果

实验拓扑



接口及 IP 地址规划

设备	接口	类型	规划
SW1	E0/0	Access	VLAN 10，连接 PC1
SW1	E0/1	Trunk	Allow VLAN 10、20
SW1	E0/2	Trunk	Allow VLAN 10、20
SW2	E0/1	Trunk	Allow VLAN 10、20
SW2	E0/2	Trunk	Allow VLAN 10、20

设备	接口	类型	规划
SW2	E0/3	Trunk	Allow VLAN 10、20
SW3	E0/1	Trunk	Allow VLAN 10、20
SW3	E0/2	Trunk	Allow VLAN 10、20
SW3	E0/3	Trunk	Allow VLAN 10、20
SW4	E0/1	Trunk	Allow VLAN 10、20
SW4	E0/2	Trunk	Allow VLAN 10、20
SW4	E0/3	Trunk	Allow VLAN 10、20
SW5	E0/1	Trunk	Allow VLAN 10、20
SW5	E0/2	Trunk	Allow VLAN 10、20
SW5	E0/3	Trunk	Allow VLAN 10、20
SW6	E0/1	Trunk	Allow VLAN 10、20
SW6	E0/2	Trunk	Allow VLAN 10、20
SW6	E0/3	Access	VLAN 10，连接 PC2
PC1	eth0	-	192.168.10.1/24
PC2	eth0	-	192.168.10.2/24

接口连接规划

设备	接口	连接设备	连接接口
PC1	eth0	SW1	E0/0
SW1	E0/1	SW2	E0/1
SW1	E0/2	SW3	E0/1
SW2	E0/2	SW3	E0/2
SW2	E0/3	SW4	E0/1
SW3	E0/3	SW5	E0/1
SW4	E0/2	SW5	E0/2
SW4	E0/3	SW6	E0/1

设备	接口	连接设备	连接接口
SW5	E0/3	SW6	E0/2
SW6	E0/3	PC2	eth0

STP 模式规划

设备	STP 模式	角色规划	说明
SW1	MST	CIST Root	MST 区域核心根桥
SW2	MST	MST 成员交换机	连接 MST 与 RSTP 区域
SW3	MST	MST 成员交换机	连接 MST 与 RSTP 区域
SW4	RSTP	RSTP 区域交换机	与 SW2 形成边界连接
SW5	RSTP	RSTP 区域交换机	与 SW3 形成边界连接
SW6	RSTP	RSTP 区域接入端	连接 PC2

MST Region 规划

设备	Region Name	Revision	MST 实例	VLAN 映射
SW1、SW2、SW3	CCNP	1	MSTI 1	VLAN 10
SW1、SW2、SW3	CCNP	1	MSTI 2	VLAN 20

CIST 端口角色预期

链路	本端角色	对端角色	说明
PC1-SW1	Edge / FWD	-	接入终端链路
SW1-SW2	SW1 DP / FWD	SW2 RP / FWD	SW2 到根桥路径
SW1-SW3	SW1 DP / FWD	SW3 RP / FWD	SW3 到根桥路径
SW2-SW3	SW2 DP / FWD	SW3 ALT / DISC	MST 区域冗余链路
SW2-SW4	SW2 DP / FWD	SW4 RP / FWD	Boundary 链路
SW3-SW5	SW3 DP / FWD	SW5 RP / FWD	Boundary 链路
SW4-SW5	SW4 DP / FWD	SW5 ALT / DISC	RSTP 冗余链路

链路	本端角色	对端角色	说明
SW4-SW6	SW4 DP / FWD	SW6 RP / FWD	SW6 到根桥路径
SW5-SW6	SW5 DP / FWD	SW6 ALT / DISC	RSTP 冗余链路
SW6-PC2	Edge / FWD	-	接入终端链路

配置需求

1. 根据实验拓扑完成设备连接与基础配置
2. 按照规划表完成 PC1、PC2 的 IP 地址配置
3. 在 SW1、SW2、SW3、SW4、SW5、SW6 上创建 VLAN 10、VLAN 20
4. 将 SW1 的 E0/0 配置为 Access 端口并加入 VLAN 10
5. 将 SW6 的 E0/3 配置为 Access 端口并加入 VLAN 10
6. 将所有交换机互联端口配置为 Trunk，并只允许 VLAN 10、VLAN 20 通过
7. 将 SW1、SW2、SW3 配置为 MST 模式，并使用相同的 Region Name、Revision 和 VLAN 映射
8. 将 SW1 配置为 CIST Root，使 SW1 成为整个混合网络的根桥
9. 调整 SW2、SW3 的 STP 优先级，使 MST 区域内端口角色符合拓扑预期
10. 将 SW4、SW5、SW6 配置为 RSTP 模式，并调整优先级使 RSTP 区域端口角色符合拓扑预期
11. 验证 SW2-SW4、SW3-SW5 是否成为 MST 与 RSTP 的边界链路
12. 验证 PC1 与 PC2 是否可以正常通信
13. 关闭一条当前转发链路，观察 STP/RSTP/MST 的端口状态变化与收敛过程

作业要求

1. 请给出 SW1、SW2、SW3、SW4、SW5、SW6 的最终配置，并按照设备进行分类

SW1

```
#SW1
SW1#SHOW RUN
Building configuration...
```

```
Current configuration : 1657 bytes
!
! Last configuration change at 21:54:57 UTC Tue Oct 13 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SW1
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode mst
spanning-tree extend system-id
!
spanning-tree mst configuration
 name CCNP
```

```
 revision 1
 instance 1 vlan 10
 instance 2 vlan 20
!
spanning-tree mst 1-2 priority 24576
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface Ethernet0/0
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast edge
!
interface Ethernet0/1
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/2
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/3
!
interface Ethernet1/0
```

```
!
interface Ethernet1/1
!
interface Ethernet1/2
!
interface Ethernet1/3
!
interface Ethernet2/0
!
interface Ethernet2/1
!
interface Ethernet2/2
!
interface Ethernet2/3
!
interface Ethernet3/0
!
interface Ethernet3/1
!
interface Ethernet3/2
!
interface Ethernet3/3
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
```

```
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
!
!
end
```

SW2

```
SW2



!
! Last configuration change at 21:56:03 UTC Tue Oct 13 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SW2
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
```

```
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode mst
spanning-tree extend system-id
!
spanning-tree mst configuration
 name CCNP
 revision 1
 instance 1 vlan 10
 instance 2 vlan 20
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
```

```
interface Ethernet0/0
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/1
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/2
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/3
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet1/0
 no shutdown
!
interface Ethernet1/1
 no shutdown
!
interface Ethernet1/2
 no shutdown
!
interface Ethernet1/3
 no shutdown
!
interface Ethernet2/0
 no shutdown
```

```
!
interface Ethernet2/1
 no shutdown
!
interface Ethernet2/2
 no shutdown
!
interface Ethernet2/3
 no shutdown
!
interface Ethernet3/0
 no shutdown
!
interface Ethernet3/1
 no shutdown
!
interface Ethernet3/2
 no shutdown
!
interface Ethernet3/3
 no shutdown
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
```

```
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
!
!
end
```

SW3

```
SW3

!
! Last configuration change at 21:56:02 UTC Tue Oct 13 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SW3
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
```

```
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode mst
spanning-tree extend system-id
!
spanning-tree mst configuration
 name CCNP
 revision 1
 instance 1 vlan 10
 instance 2 vlan 20
!
!
!
```

```
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/1
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/2
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/3
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet1/0
 no shutdown
!
interface Ethernet1/1
 no shutdown
!
interface Ethernet1/2
 no shutdown
!
interface Ethernet1/3
 no shutdown
!
interface Ethernet2/0
 no shutdown
!
```

```
interface Ethernet2/1
 no shutdown
!
interface Ethernet2/2
 no shutdown
!
interface Ethernet2/3
 no shutdown
!
interface Ethernet3/0
 no shutdown
!
interface Ethernet3/1
 no shutdown
!
interface Ethernet3/2
 no shutdown
!
interface Ethernet3/3
 no shutdown
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
```

```
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
!
!
end
```

SW4

```
SW4

!
! Last configuration change at 21:56:01 UTC Tue Oct 13 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SW4
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
```

```
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface Ethernet0/0
 no shutdown
!
interface Ethernet0/1
 no shutdown
!
interface Ethernet0/2
 no shutdown
```

```
!
interface Ethernet0/3
 no shutdown
!
interface Ethernet1/0
 no shutdown
!
interface Ethernet1/1
 no shutdown
!
interface Ethernet1/2
 no shutdown
!
interface Ethernet1/3
 no shutdown
!
interface Ethernet2/0
 no shutdown
!
interface Ethernet2/1
 no shutdown
!
interface Ethernet2/2
 no shutdown
!
interface Ethernet2/3
 no shutdown
!
interface Ethernet3/0
 no shutdown
!
interface Ethernet3/1
 no shutdown
!
interface Ethernet3/2
 no shutdown
!
interface Ethernet3/3
```

```
 no shutdown
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
!
!
end
```

SW5

```
SW5
```

```
!
! Last configuration change at 21:56:00 UTC Tue Oct 13 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SW5
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
!
!
!
```

```
!
!
!
!
!
!
!
!
!
!
!
interface Ethernet0/0
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/1
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/2
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/3
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet1/0
 no shutdown
!
```

```
interface Ethernet1/1
 no shutdown
!
interface Ethernet1/2
 no shutdown
!
interface Ethernet1/3
 no shutdown
!
interface Ethernet2/0
 no shutdown
!
interface Ethernet2/1
 no shutdown
!
interface Ethernet2/2
 no shutdown
!
interface Ethernet2/3
 no shutdown
!
interface Ethernet3/0
 no shutdown
!
interface Ethernet3/1
 no shutdown
!
interface Ethernet3/2
 no shutdown
!
interface Ethernet3/3
 no shutdown
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
```

```
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
line vty 0 4
 login
!
!
!
end
```

SW6

```
SW6

!
! Last configuration change at 21:55:59 UTC Tue Oct 13 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
```

```
service compress-config
!
hostname SW6
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
!
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
!
interface Ethernet0/0
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/1
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/2
 no shutdown
 switchport trunk allowed vlan 10,20
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/3
 no shutdown
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast edge
!
interface Ethernet1/0
 no shutdown
!
interface Ethernet1/1
 no shutdown
!
interface Ethernet1/2
 no shutdown
!
interface Ethernet1/3
```

```
 no shutdown
!
interface Ethernet2/0
 no shutdown
!
interface Ethernet2/1
 no shutdown
!
interface Ethernet2/2
 no shutdown
!
interface Ethernet2/3
 no shutdown
!
interface Ethernet3/0
 no shutdown
!
interface Ethernet3/1
 no shutdown
!
interface Ethernet3/2
 no shutdown
!
interface Ethernet3/3
 no shutdown
!
ip forward-protocol nd
!
ip tcp synwait-time 5
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
```

```
 control-plane
 !
 !
 line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
 line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
 line vty 0 4
  login
 !
 !
 !
 end
```

2. 请给出 PC1、PC2 的 IP 地址配置结果

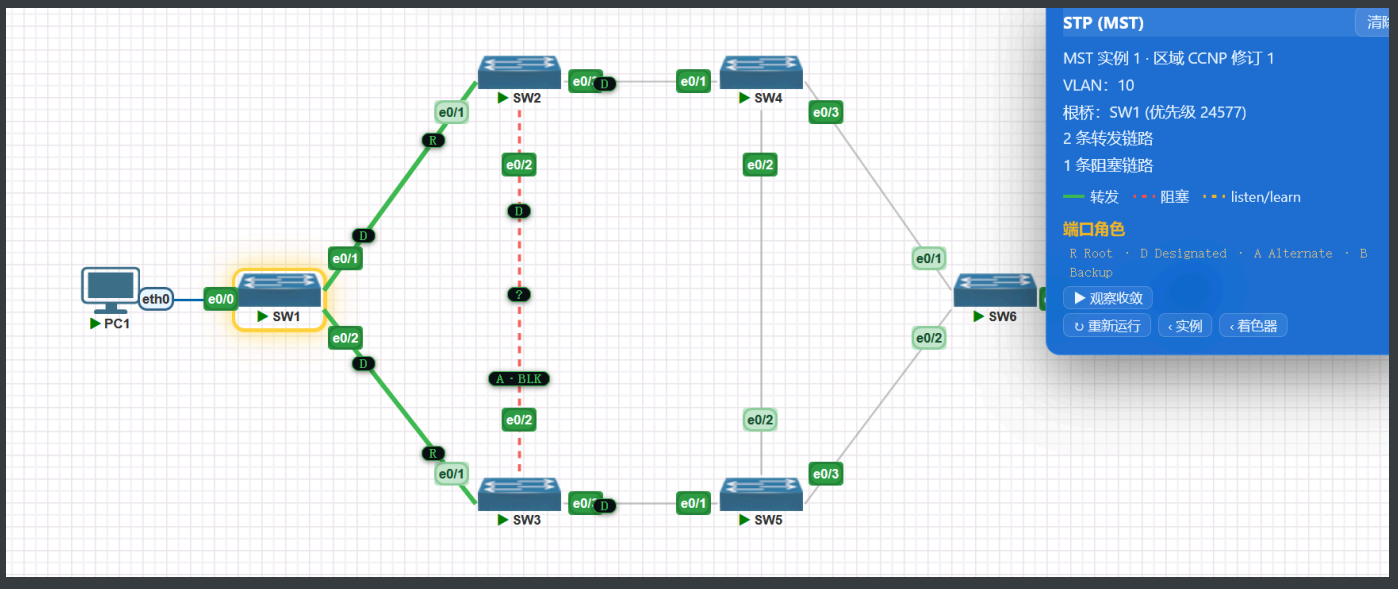
```
PC1> show ip

NAME        : PC1[1]
IP/MASK     : 192.168.10.1/24
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:07
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500
```

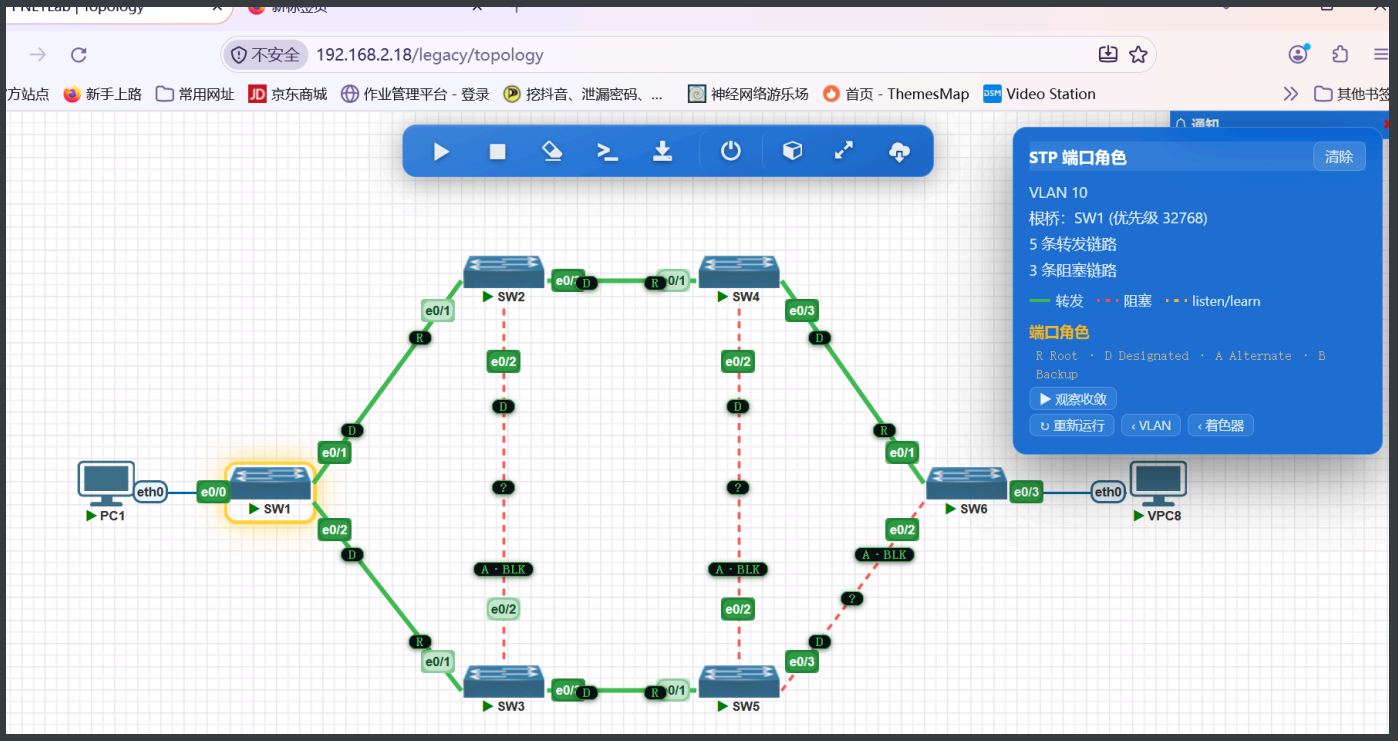
```
PC2> show ip

NAME        : PC2[1]
IP/MASK     : 192.168.10.2/24
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:0e
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500
```

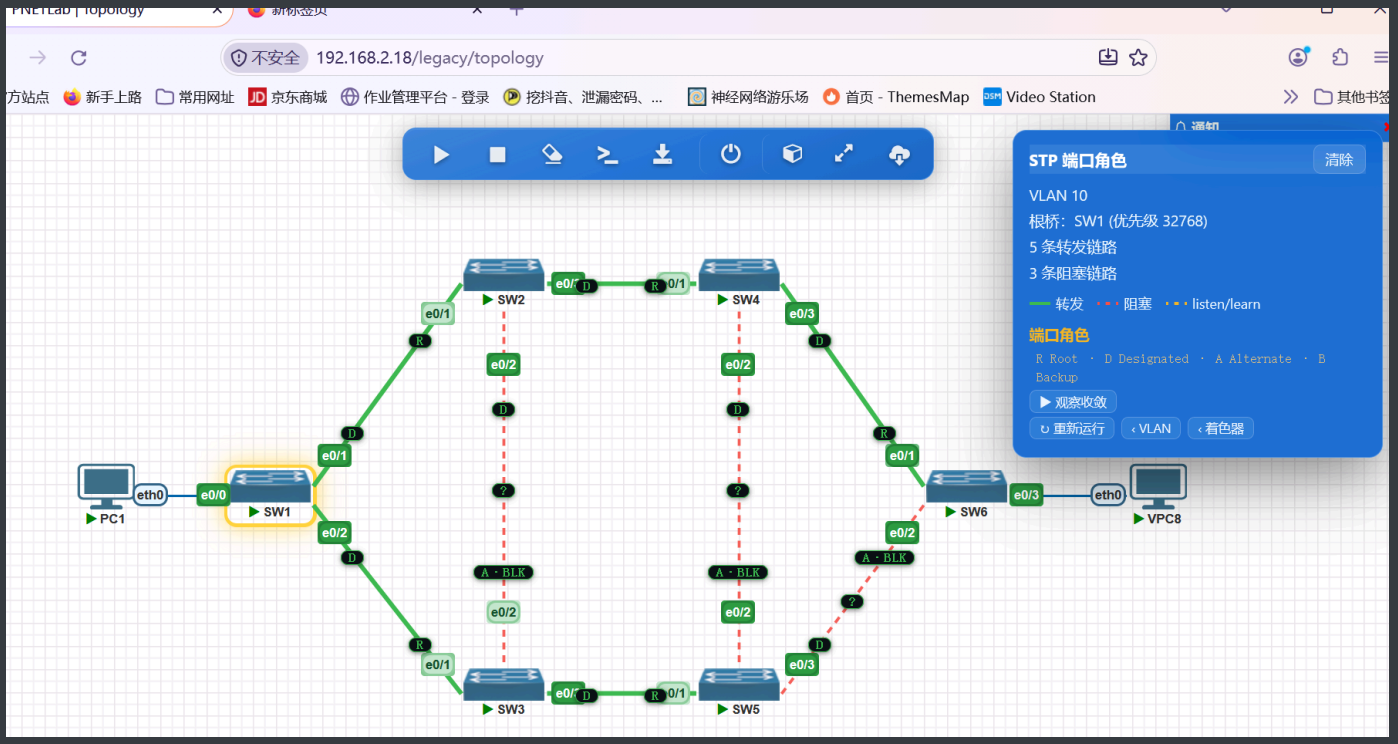
3. 请给出 MST Region 配置验证截图



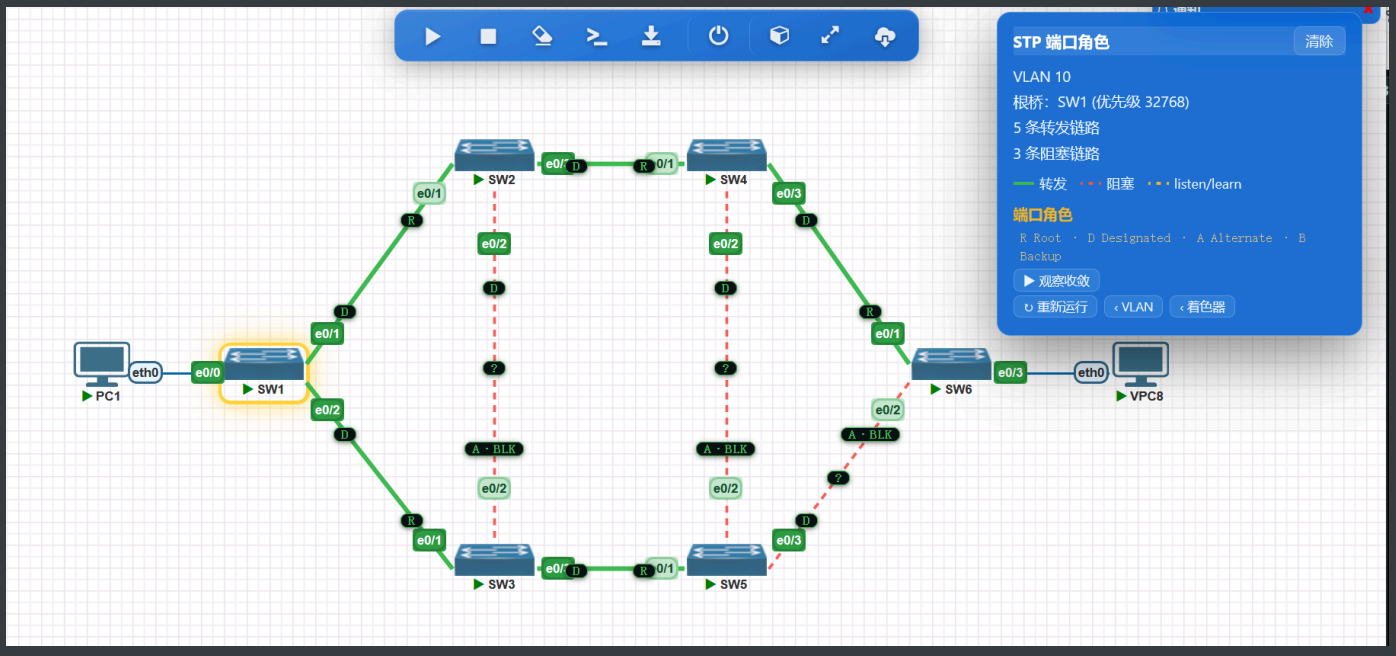
4. 请给出 CIST Root 验证截图



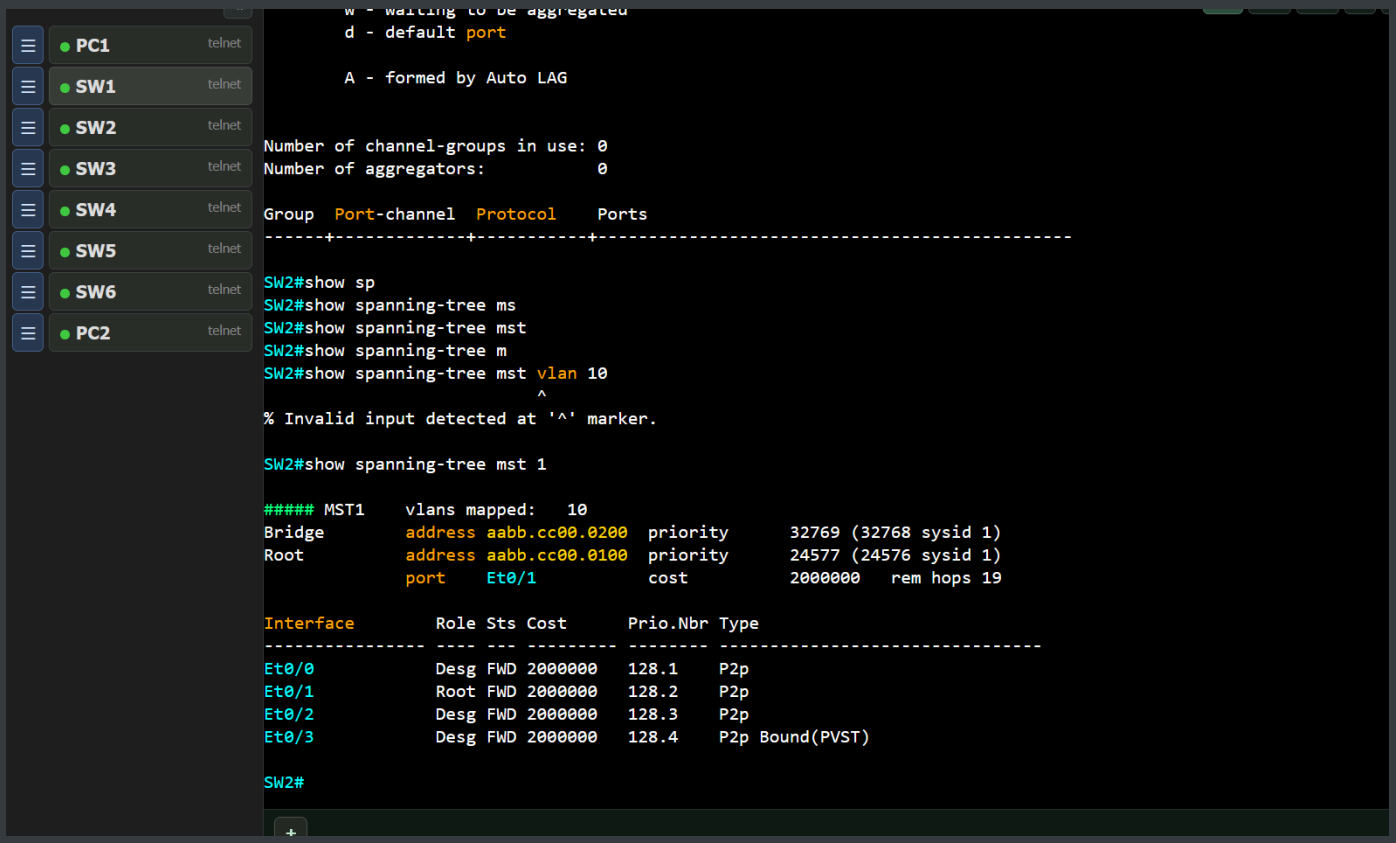
5. 请给出各关键链路端口角色与端口状态的验证截图



6. 请给出 SW2-SW4、SW3-SW5 边界端口的验证截图



SW2-SW4



```
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 0
Number of aggregators:           0

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------------------------

SW2#show sp
SW2#show spanning-tree ms
SW2#show spanning-tree mst
SW2#show spanning-tree m
SW2#show spanning-tree mst vlan 10
                           ^
% Invalid input detected at '^' marker.

SW2#show spanning-tree mst 1

##### MST1    vlans mapped:   10
Bridge        address aabb.cc00.0200  priority      32769 (32768 sysid 1)
Root          address aabb.cc00.0100  priority      24577 (24576 sysid 1)
              port    Et0/1           cost          2000000   rem hops 19

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Et0/0            Desg FWD 2000000   128.1    P2p
Et0/1            Root FWD 2000000   128.2    P2p
Et0/2            Desg FWD 2000000   128.3    P2p
Et0/3            Desg FWD 2000000   128.4    P2p Bound(PVST)

SW2#
```

```
SW4#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID     Priority    32768
              Address     aabb.cc00.0100
              Cost        100
              Port        2 (Ethernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32778  (priority 32768 sys-id-ext 10)
              Address     aabb.cc00.0400
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/1               Root FWD 100       128.2    P2p Peer(STP)
Et0/2               Desg FWD 100       128.3    P2p
Et0/3               Desg FWD 100       128.4    P2p


SW4#
```

SW3-SW5

```
SW3#show spanning-tree mst 1

##### MST1    vlans mapped:   10
Bridge        address aabb.cc00.0300  priority      32769 (32768 sysid 1)
Root          address aabb.cc00.0100  priority      24577 (24576 sysid 1)
              port    Et0/1           cost          2000000   rem hops 19

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Et0/0            Desg FWD 2000000   128.1    P2p
Et0/1            Root FWD 2000000   128.2    P2p
Et0/2            Altn BLK 2000000   128.3    P2p
Et0/3            Desg FWD 2000000   128.4    P2p Bound(PVST)

SW3#
```

```
SW5#show sp
SW5#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID     Priority    32768
              Address     aabb.cc00.0100
              Cost        100
              Port        2 (Ethernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32778  (priority 32768 sys-id-ext 10)
              Address     aabb.cc00.0500
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 100       128.1    P2p
Et0/1               Root FWD 100       128.2    P2p Peer(STP)
Et0/2               Altn BLK 100       128.3    P2p
Et0/3               Desg FWD 100       128.4    P2p


SW5#
```

7. 请给出 PC1 与 PC2 连通性验证截图

```
VPC1> ping 192.168.10.2 -t

84 bytes from 192.168.10.2 icmp_seq=1 ttl=64 time=0.466 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=64 time=1.071 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=64 time=0.649 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=64 time=1.947 ms
```

8. 请说明 SW1 为什么会成为 CIST Root

   答：因为在配置sw1时，设置root primary ，桥优先级最高，而对于整个cist中，sw1的桥优先级也是最高的，所以cist下发的最优的根桥是sw1的桥id

9. 请说明 SW2-SW4、SW3-SW5 为什么是 Boundary 链路

   答：因为sw1-3是mst， SW2-SW4、SW3-SW5是通向外部的边界端口的按时